

JORDAN B. S. VAN BEECK

Asteroseismology | Stellar Astrophysics | FOSS



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SCIENTIFIC EDUCATION

Ph.D., Astronomy & Astrophysics, KU Leuven (Belgium), September 2023

Asteroseismology of Kepler B stars: internal magnetism and non-linear mode coupling

Promoters: Prof. Dr. C. Aerts, Prof. Dr. T. Van Hoolst, Prof. Dr. D. M. Bowman

M.Sc., Astronomy & Astrophysics, KU Leuven (Belgium), July 2019 – *Magna Cum Laude*

The influence of an interior magnetic field on gravity-mode oscillations of intermediate-mass stars

Promoters: Prof. Dr. C. Aerts, Dr. T. Van Reeth, Prof. Dr. D. M. Bowman

M.Sc., Chemistry, University of Antwerp (Belgium), September 2017 – *Magna Cum Laude*

Characterization of radioactive particles

Promoters: Prof. Dr. K. Janssens, Prof. Dr. B. Salbu, Prof. Dr. O.-C. Lind, M.Sc. G. Nuyts

B.Sc., Chemistry, University of Antwerp (Belgium), September 2015 – *Magna Cum Laude*

Atomic scale reactive MD studies of DNA oxidation for plasma oncology: the role of H₂O₂ and HO₂

Promoters: Prof. Dr. A. Bogaerts, Prof. Dr. E. Neyts, Dr. C. Verlackt

EMPLOYMENT

Heidelberg Institute for Theoretical Studies (HITS), TOS group, Heidelberg, Germany

Postdoctoral Researcher Asteroseismology February 2025 – March 2027

Project: higher-order rotational effects on mixed-mode oscillations of red giants

KU Leuven, Department of Physics and Astronomy, Institute of Astronomy, Leuven, Belgium

Research Associate, Asteroseismology, September 2023 – December 2023

Project: non-linear asteroseismic modeling of slowly pulsating B stars

AWARDS AND FELLOWSHIPS

2021 - 2022	FWO(Research Foundation Flanders) long research stay grant (€12 738) for a stay as a VSR at TAPIR, Caltech, CA, USA with Prof. Dr. J. Fuller.
2019 - 2023	4-year PhD Fellowship, Department of Physics and Astronomy, KU Leuven
2019	Paul Smeyers Prize, KU Leuven Awarded to the annual best master's thesis in Astronomy & Astrophysics at the June examination session.
2016	Erasmus+ M.Sc. internship for a stay at CERAD, NMBU, Norway.

TEACHING EXPERIENCE

Lecturer, Heidelberg University

2025	Theoretical introduction for 'Asteroseismology' (Master-level physics course that introduces various aspects of asteroseismology.)
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Teaching Assistant, Department of Physics and Astronomy, KU Leuven

2022 - 2023	K01B4A: 'Natuurkunde met elementen van wiskunde' I (taught in Dutch) (Introductory physics course for the first-year students in the B.Sc. of pharmaceutical sciences)
2019 - 2021	K01B4A B: 'Natuurkunde met elementen van wiskunde' I II (taught in Dutch) (Introductory physics courses for the first-year students in the B.Sc. of pharmaceutical sciences) English translation of course titles: 'Physics with elements of mathematics' I II

VISITING RESEARCH POSITIONS

11/2021 - 06/2022	Visiting Student Researcher (VSR) at TAPIR, Caltech, CA, USA Project goal: development of computational tools that aid in explaining oscillation mode amplitude limitation in slowly pulsating B stars Mentor: Prof. Dr. J. Fuller <i>Funding: FWO long research stay grant</i>
11/2021 - 12/2021	Transport in Stellar Interiors Program Affiliate (workshop, conference) at KITP, UC Santa Barbara, CA, USA Mentor: Prof. Dr. L. Bildsten <i>Funding: FWO long research stay grant</i>
04/2016	Visiting Student Researcher at DESY, Hamburg, Germany Project goal: high spatial and spectral resolution X-ray analysis of environmental radioactive particles Mentors: Prof. Dr. Ole-Christian Lind, Prof. Dr. Koen Janssens, M.Sc. Gert Nuyts
01/2016 - 06/2016	Erasmus+ internship at CERAD, NMBU, Norway Project goal: Characterization of environmental radioactive particles Mentor: Prof. Dr. Ole-Christian Lind <i>Funding: Erasmus+ M.Sc. internship grant</i>
05/2015	Visiting Student Researcher at ESRF, Grenoble, France Project goal: high spatial and spectral resolution X-ray analysis of art samples Mentors: Dr. Wout De Nolf, M.Sc. Gert Nuyts

I was briefly a student researcher at the Institute of Astronomy, KU Leuven, Belgium from 02/2018 until 05/2018 that worked on the simulation of dust cluster nucleation in carbon-rich winds of asymptotic giant branch stars. This work was performed under the mentorship of Dr. David Gobrecht and Prof. Dr. Leen Decin and (indirectly) resulted in a contribution to a published technical report.

SEMINARS/CONFERENCE PRESENTATIONS

July 2026	POSTER PRESENTATION (doi:10.5281/zenodo.21083361) TASC10 / KASC17 workshop/conference at Aarhus University, Denmark.
July 2025	CONTRIBUTED TALK (doi:10.5281/zenodo.17241657) TASC9 / KASC16 workshop/conference at ISTA, Austria.
March 2023	INVITED SEMINAR (https://youtu.be/dWaEkri_s2I?feature=shared) The Good Vibrations seminar series, S3E8.
February 2023	INVITED SEMINAR (https://youtu.be/nTWZoLCMa_g?feature=shared) Ad-valvas seminar of the Institute of Astronomy, KU Leuven, Belgium.
July 2022	POSTER PRESENTATION (additional info: doi:10.5281/zenodo.6814504) TASC6 / KASC13 workshop/conference at KU Leuven, Belgium.
November 2021	CONTRIBUTED TALK (doi:10.26081/K6VH15) Transfer in Stellar Interiors conference (part of the Probes of Transport in Stars program), Kavli Institute for Theoretical Physics (KITP), UC Santa Barbara, CA, USA.
July 2020	CONTRIBUTED TALK (doi:10.5281/zenodo.5525346) MOBSTER-1 Virtual conference 2020: Stellar variability as a probe of magnetic fields in massive stars (online, hosted by University of Delaware, DE, USA)
June 2019	POSTER PRESENTATION (doi:10.5281/zenodo.3228857) 74th Dutch Astronomers Conference/Nederlandse Astronomenconferentie Kapteyn Astronomical Institute, University of Groningen, Groningen/Paterswolde, the Netherlands.

PROFESSIONAL ACTIVITIES

Thematic research school/workshop participation

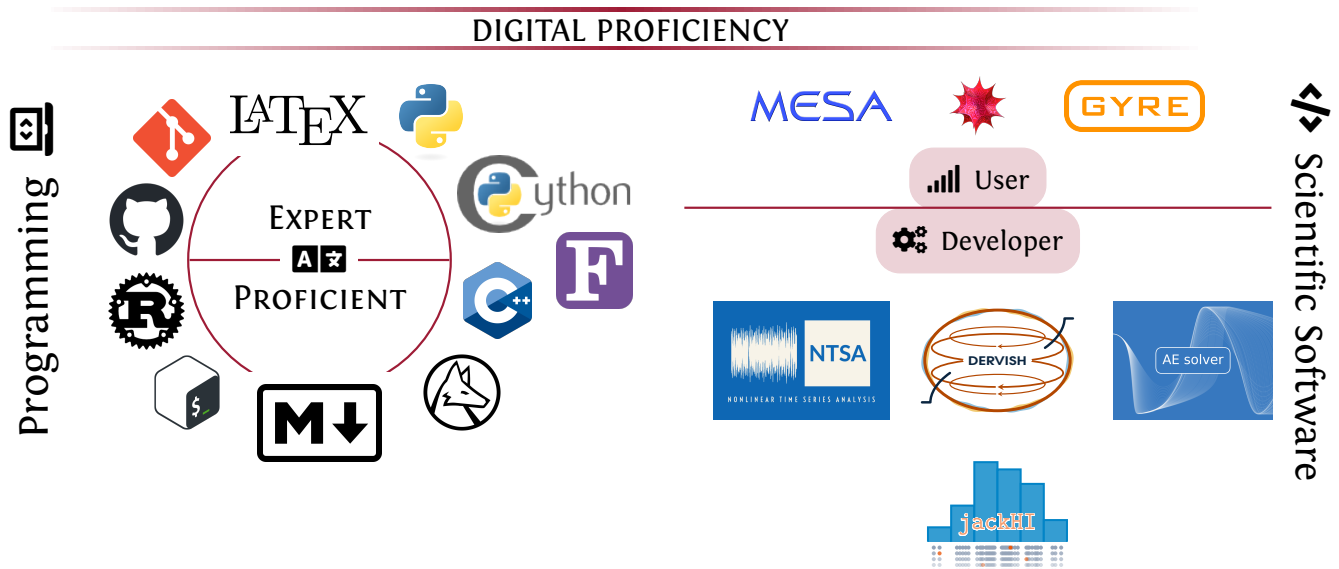
- April 2026 Phlegethon ([doi:10.5281/zenodo.19554676](https://doi.org/10.5281/zenodo.19554676)) workshop, HITS, Heidelberg, Germany
- December 2025 XIX Winter Workshop on Stellar Astrophysics 2025, HITS, Heidelberg, Germany
- Nov.-Dec. 2021 Probes of Transport in Stars program, KITP, UC Santa Barbara, CA, USA
- August 2021 10th MESA summer school (online: <https://mesastar.org/summer-school-2021/>)
- July 2020 Let's Talk Science: 8th Summer School for Science Communication and Communicative Competences
- April 2019 STFC/MAMSIE mini-workshop
Workshop facilitating research collaborations among the KU Leuven MAMSIE group led by [Prof. Dr. Conny Aerts](#), the research group at Newcastle University led by [Prof. Dr. Tamara Rogers](#), and the research group at CEA led by [Dr. Stéphane Mathis](#).
- October 2018 STFC/MAMSIE mini-workshop
Workshop facilitating research collaborations among the KU Leuven MAMSIE group led by [Prof. Dr. Conny Aerts](#), the research group at Newcastle University led by [Prof. Dr. Tamara Rogers](#), and the research group at CEA led by [Dr. Stéphane Mathis](#).

Conference organization

- July 2022 Member of the Local Organizing Committee
TASC6 / KASC13 workshop/conference in Leuven, Belgium

Professionally associated as

- 2026 – ... Junior Member of the [International Astronomical Union \(IAU\)](#)
- 2022 Fellow (Graduate Student) of the [Royal Astronomical Society \(RAS\)](#)
- 2020 – 2022 Graduate Student Member of the [American Astronomical Society \(AAS\)](#)
- 2020 – 2022 Graduate Student Member of the [Royal Netherlands Astronomical Society \(KNA\)](#)
- 2019 – 2023 Graduate Student Member of the [International Research Network for Nuclear Astrophysics \(IReNA\)](#)



List of Publications

CORRESPONDING AUTHOR PEER-REVIEWED PUBLICATIONS

1. **J. Van Beeck**, T. Van Hoolst, C. Aerts, and J. Fuller. Non-linear Three-mode Coupling of Gravity Modes in Rotating Slowly Pulsating B Stars: Stationary Solutions and Modeling Potential. *A & A*, volume 687, article id. A265, July 2024. (DOI: [10.1051/0004-6361/202348369](https://doi.org/10.1051/0004-6361/202348369)).
Accompanying Zenodo page: <https://zenodo.org/records/10814654>.
Uses the [AE Solver](#) framework and light curve models generated by [NTSA](#).
2. **J. Van Beeck**, D. M. Bowman, M. G. Pedersen, T. Van Reeth, T. Van Hoolst, and C. Aerts. Detection of non-linear resonances among gravity modes of slowly pulsating B stars: Results from five iterative pre-whitening strategies. *A & A*, volume 655, article id. A59, November 2021. Uses the [NTSA](#) framework to generate the light curve models and resonance signatures. (DOI: [10.1051/0004-6361/202141572](https://doi.org/10.1051/0004-6361/202141572))
3. **J. Van Beeck**, V. Prat, T. Van Reeth, S. Mathis, D. M. Bowman, C. Neiner, and C. Aerts. Detecting axisymmetric magnetic fields using gravity modes in intermediate-mass stars. *A & A*, volume 638, article id. A149, June 2020. (DOI: [10.1051/0004-6361/201937363](https://doi.org/10.1051/0004-6361/201937363); Inlists: Downloadable from [the MESA marketplace](#) or from DOI: [10.5281/zenodo.3818245](https://doi.org/10.5281/zenodo.3818245))

CO-AUTHORED PEER-REVIEWED PUBLICATIONS

1. A. A. Patil, C. Aerts, N. Y. N. Wang, **J. Van Beeck**, and M. G. Pedersen. Beyond prewhitening: detection of gravity modes and their period spacings in slowly pulsating B stars using the multitaper F-test. *Accepted for publication in A & A*.
My Contributions: Provide SPB star output data obtained for [Van Beeck et al. \(2021\)](#), check equations, provide extensive feedback on entire manuscript.
2. J. Müller, Q. Coppée, **J. Van Beeck**, T. van Lier, and S. Hekker. Asymptotic power spectra and visibilities of damped mixed modes. *A & A*, volume 707, article id. A126, March 2026. (DOI: [10.1051/0004-6361/202557877](https://doi.org/10.1051/0004-6361/202557877))
My Contributions: Check derivations, discuss initial setup of theoretical framework and provide extensive feedback on entire manuscript.
3. F. Ahlborn, J. J. M. Ong, **J. Van Beeck**, E. P. Bellinger, S. Hekker, and S. Basu. Impact of near-degeneracy effects on linear rotational inversions for red giant stars. *A & A*, volume 704, article id. A230, December 2025. (DOI: [10.1051/0004-6361/20255537](https://doi.org/10.1051/0004-6361/20255537))
My Contributions: Check derivations, write Appendices A, C (developing a generalized form of the near-degeneracy formalism of [Dziembowski and Goode \(1992\)](#)), and D, write the generalized NDE QEP solver (Python), and provide extensive feedback on entire manuscript.
4. D.J. Fritzewski, T. Van Reeth, C. Aerts, **J. Van Beeck**, S. Gossage, G. Li. Age-dating the young open cluster UBC1 with g-mode asteroseismology, gyrochronology, and isochrone fitting. *A & A*, volume 681, article id. A13, January 2024. (DOI: [10.1051/0004-6361/202347618](https://doi.org/10.1051/0004-6361/202347618))
My Contributions: make [NTSA](#) available for the lead author; provide guidance in using the automated [NTSA](#) framework used to generate the light curve models, provide feedback on entire manuscript.
5. T. Van Reeth, C. Johnston, J. Southworth, J. Fuller, D. M. Bowman, L. Poniowski, and **J. Van Beeck**. Tidally perturbed gravity-mode pulsations in a sample of close eclipsing binaries. *A & A*, volume 671, article id. A121, March 2023. (DOI: [10.1051/0004-6361/202245460](https://doi.org/10.1051/0004-6361/202245460))
My Contributions: provide the automated [NTSA](#) framework used to generate the light curve model, check derivations for formalism, provide feedback on entire manuscript.
6. T. Van Reeth, P. De Cat, **J. Van Beeck**, V. Prat, D. J. Wright, H. Lehmann, A.-N. Chené, E. Kambe, S. L. S. Yang, G. Gentile and M. Joos. The near-core rotation of HD 112429. A γ Doradus star with TESS photometry and legacy spectroscopy. *A & A*, volume 662, article id. A58, June 2022. (DOI: [10.1051/0004-6361/202142921](https://doi.org/10.1051/0004-6361/202142921))
My Contributions: provide the automated [NTSA](#) framework used to generate the light curve and LSD profile variations model, check derivations for formalism in Appendix A, provide feedback on entire manuscript.
7. T. Van Reeth, J. Southworth, **J. Van Beeck**, and D. M. Bowman. V456 Cyg: An eclipsing binary with tidally perturbed g-mode pulsations. *A & A*, volume 659, article id. A177, March 2022. (DOI: [10.1051/0004-6361/202142833](https://doi.org/10.1051/0004-6361/202142833))
My Contributions: make [NTSA](#) available for the lead author; provide guidance in using the automated [NTSA](#) framework used to generate the pulsation light curve model, provide feedback on entire manuscript.

8. C. Aerts, K. Augustson, S. Mathis, M. G. Pedersen, J. S. G. Mombarg, V. Vanlaer, **J. Van Beeck**, and T. Van Reeth. Rossby numbers and stiffness values inferred from gravity-mode asteroseismology of rotating F- and B-type dwarfs. Consequences for mixing, transport, magnetism, and convective penetration. *A & A*, volume 656, article id. A121, December 2021. (DOI: [10.1051/0004-6361/202142151](https://doi.org/10.1051/0004-6361/202142151))
My Contributions: provide residual light curves based on NTSA light curve models for 26 SPB stars, compute residual-to-coherent amplitude ratios for 17 SPB stars, provide feedback on entire manuscript.
9. V. Prat, S. Mathis, C. Neiner, **J. Van Beeck**, D. M. Bowman, and C. Aerts. Period spacing of gravity modes in rapidly rotating magnetic stars. II. The case of an oblique dipolar fossil magnetic field. *A & A*, volume 636, article id. A100, April 2020. (DOI: [10.1051/0004-6361/201937398](https://doi.org/10.1051/0004-6361/201937398))
My Contributions: check derivations and code for formalism, provide feedback on entire manuscript (this article used the stellar evolution and oscillation/pulsation models computed for Prat et al. 2019).
10. V. Prat, S. Mathis, B. Buysschaert, **J. Van Beeck**, D. M. Bowman, C. Aerts, and C. Neiner. Period spacings of gravity modes in rapidly rotating magnetic stars I. Axisymmetric fossil field with poloidal and toroidal components. *A & A*, Volume 627, article id. A64, July 2019. (DOI: [10.1051/0004-6361/201935462](https://doi.org/10.1051/0004-6361/201935462))
My Contributions: compute stellar evolution and pulsation/oscillation models for the frequency perturbation calculations, check derivations and code for formalism, provide feedback on entire manuscript.

CONFERENCE PROCEEDINGS / ABSTRACTS

1. **J. Van Beeck**, T. Van Hoolst, J. Fuller, and C. Aerts. Resonant Amplitude Equations for Gravito-Inertial Modes: Non-linear mode coupling in a slowly pulsating B star. *TASC9/KASC16 Kepler Asteroseismic Science Consortium Workshop*, held in at the Institute of Science and Technology Austria (ISTA) on July 7-11, 2025 (session 17), id. 70. (DOI: [10.5281/zenodo.17241657](https://doi.org/10.5281/zenodo.17241657))
2. **J. Van Beeck**, V. Prat, T. Van Reeth, S. Mathis, D. M. Bowman, C. Neiner, and C. Aerts. Linking detected gravity modes to axisymmetric internal magnetic fields. *MOBSTER-1 virtual conference: Stellar variability as a probe of magnetic fields in massive stars*, Proceedings of the MOBSTER-1 virtual conference held 12-17 July 2020, id.13. (<https://ui.adsabs.harvard.edu/abs/2021mobs.confE..13V/abstract>, DOI: [10.5281/zenodo.5525345](https://doi.org/10.5281/zenodo.5525345))
3. V. Prat, S. Mathis, B. Buysschaert, **J. Van Beeck**, D. M. Bowman, C. Aerts, and C. Neiner. Effect of the magnetic field on period spacings of gravity modes in rapidly rotating stars. *Proceedings of the conference Stars and their Variability Observed from Space*, held in Vienna on August 19-23, 2019. Eds.: C. Neiner, W. W. Weiss, D. Baade, R. E. Griffin, C. C. Lovekin, A. F. J. Moffat. University of Vienna, 2020, pp.105-106. (<https://ui.adsabs.harvard.edu/abs/2020svos.conf.105P/abstract>)

TECHNICAL REPORTS

1. D. Gobrecht, S. T. Bromley, and **J. Van Beeck**. Structural crossover of small (SiC)_n clusters. HPC-Europe-3 report, August 2019.

POSTERS

1. **J. Van Beeck**, F. Ahlborn, B. Bordadágua, and S. Hekker. Dervish: second-order rotational perturbations to mixed modes. *TASC10/KASC17 Kepler Asteroseismic Science Consortium Workshop*, held at the University of Aarhus, Denmark on July 6-10, 2026. (DOI: [10.5281/zenodo.21083361](https://doi.org/10.5281/zenodo.21083361))
2. F. Ahlborn, **J. Van Beeck**, and S. Hekker. Beyond linear rotational inversions. *TASC10/KASC17 Kepler Asteroseismic Science Consortium Workshop*, held at the University of Aarhus, Denmark on July 6-10, 2026. (DOI: [10.5281/zenodo.21282444](https://doi.org/10.5281/zenodo.21282444))
3. F. Ahlborn, J. Ong, **J. Van Beeck**, E. Bellinger, S. Hekker, and S. Basu. Estimating internal rotation rates of red giants in the presence of interacting mixed modes. *TASC9/KASC16 Kepler Asteroseismic Science Consortium Workshop*, held at the Institute of Science and Technology Austria (ISTA) on July 7-11, 2025, id. 94. (DOI: [10.5281/zenodo.17432360](https://doi.org/10.5281/zenodo.17432360))
4. **J. Van Beeck**, V. Prat, S. Mathis, C. Aerts, D. Bowman, and T. Van Reeth. (2019). “Constraining magnetic fields in intermediate-mass main-sequence stars with asteroseismology.” Presented at the 74th Dutch Astronomy Conference/Nederlandse Astronomenconferentie, Groningen/Paterswolde, the Netherlands. (DOI: [10.5281/zenodo.3228857](https://doi.org/10.5281/zenodo.3228857))
5. **J. Van Beeck**, T. Van Hoolst, J. Fuller, and C. Aerts. (July 2022). “Resonant Amplitude Equations for Gravito-Inertial Modes: Mode Coupling in a Slowly Pulsating B star.” Presented at the TASC6/KASC13 workshop/conference at Leuven, Belgium. (additional information DOI: [10.5281/zenodo.6814504](https://doi.org/10.5281/zenodo.6814504), PDF: <https://fys.kuleuven.be/ster/events/conferences/2020/tasc6/posters/jordan-van-beeck-poster-1.pdf>)

(ASTRONOMY) SOFTWARE DEVELOPMENT

1. **J. Van Beeck.** jackHI: a software suite that contains functions to aid the creation of data-driven histograms.
(2026 – ...)
In active development: alpha version available as the  repository: <https://github.com/JVB11/jackHI>. Alpha version documentation available at the webpage: <https://jvb11.github.io/jackHI/>.
2. **J. Van Beeck.** Dervish: a software suite that computes up to the second order perturbation of oscillation modes due to rotation, including first-order perturbations of the eigenfunctions.
(2025 – ...)
In active development: most recent version in a private  repository; in preparation for public release (public  repository is a work in progress). Alpha version available upon reasonable request to the author.
3. **J. Van Beeck.** AE Solver – Amplitude Equation Solver: a software suite to obtain AE solutions for non-linear asteroseismology. Derives observables of non-linear mode coupling based on stellar and oscillation models.
(2021 – ...)
In active development: most recent version in a private  repository; in preparation for public release (current public  repository (work in progress): <https://github.com/JVB11/AESolver>). Beta version available upon reasonable request to the author.
4. **J. Van Beeck.** NTSA – Non-Linear Time Series Analysis: a software suite for the analysis of variability in stellar light curves. Derives signatures of non-linear oscillation mode coupling.
(2020 – ...)
In active development: most recent version in a private  repository; in preparation for public release (current public  repository (work in progress): <https://github.com/JVB11/NTSA>). Beta version available upon reasonable request to the author.
5. **J. Van Beeck and S. Nys.** RR-Lyr-Method-Comparison: Period-Metallicity-Luminosity relation validation for RR Lyrae stars (Version 1.1). (DOI: [10.5281/zenodo.2525594](https://doi.org/10.5281/zenodo.2525594))
(2018)
Not in active development; most recent version available at the  repository: <https://github.com/JVB11/RR-Lyr-Method-Comparison>. Accompanying Zenodo page: <https://zenodo.org/records/2525594>.